CS & IT ENGINEERING



Algorithms

Greedy Method

Lecture No.- 07



By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic

MCST
- Prims
- Kruskal

PYQs

Dijkstra

Topics to be Covered



- Topic
- Topic
- Topic

Questions/PYQs on MCST
Prim's Algo Code
Shortest Path Algos



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

[NAT]

P48

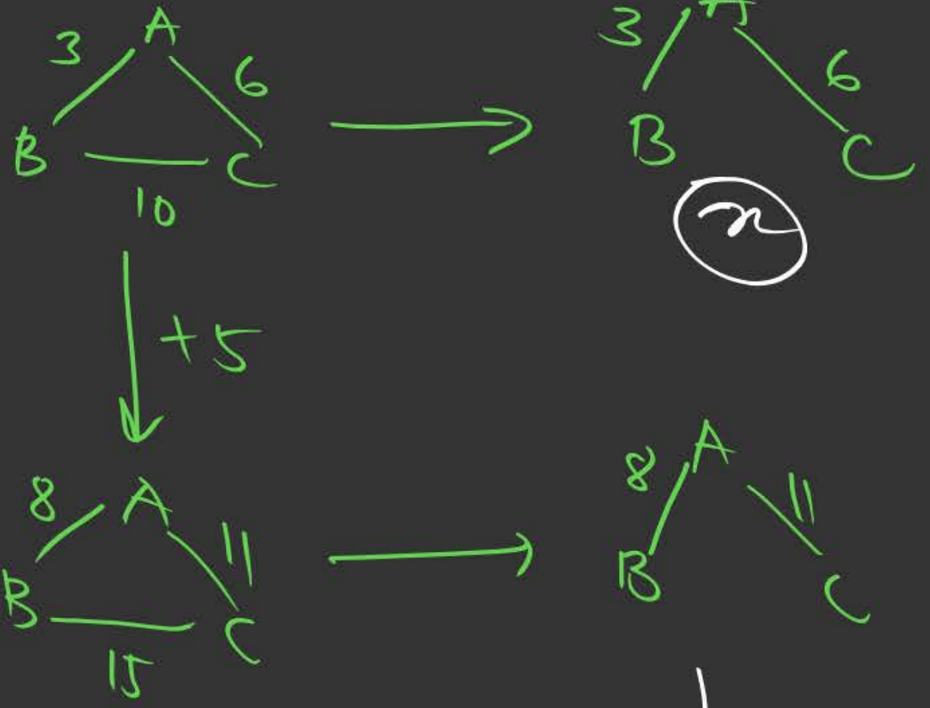
#Q. Let G a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.

+5

771

Soln :- $n = 100$

$e = 300$



$$\text{new MCST cost} = \underbrace{\text{prev MCST}}_{\text{prev MCST}} + \underbrace{(n-1)*5}_{(n-1)*5}$$

$$= 500 + (100-1) \times 5$$

$$= 500 + 99 \times 5$$

$$= \boxed{995}$$

[MCQ]

#Q. Consider a graph with 'n' vertices n > 2. The vertices are numbered V_1 to V_n. Two vertices V_i and V_j are adjacent iff $0 < |i - j| \leq 2$. The weight of such an edge is i + j. The weight of minimum cost spanning tree of such a graph for a value of n is _____.

A $n^2 - n$

B $n^2 + n + 1$

C $n^2 - n - 1$

D $n^2 - n + 1$

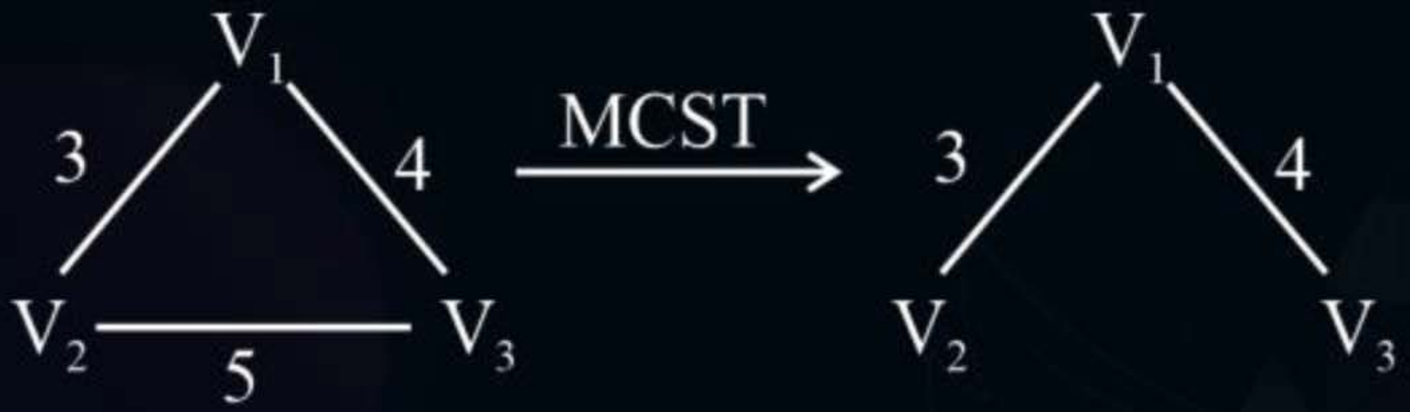


Topic : Greedy Algorithm

Solution:

Exam Appr (shortcut)

1) n = 3

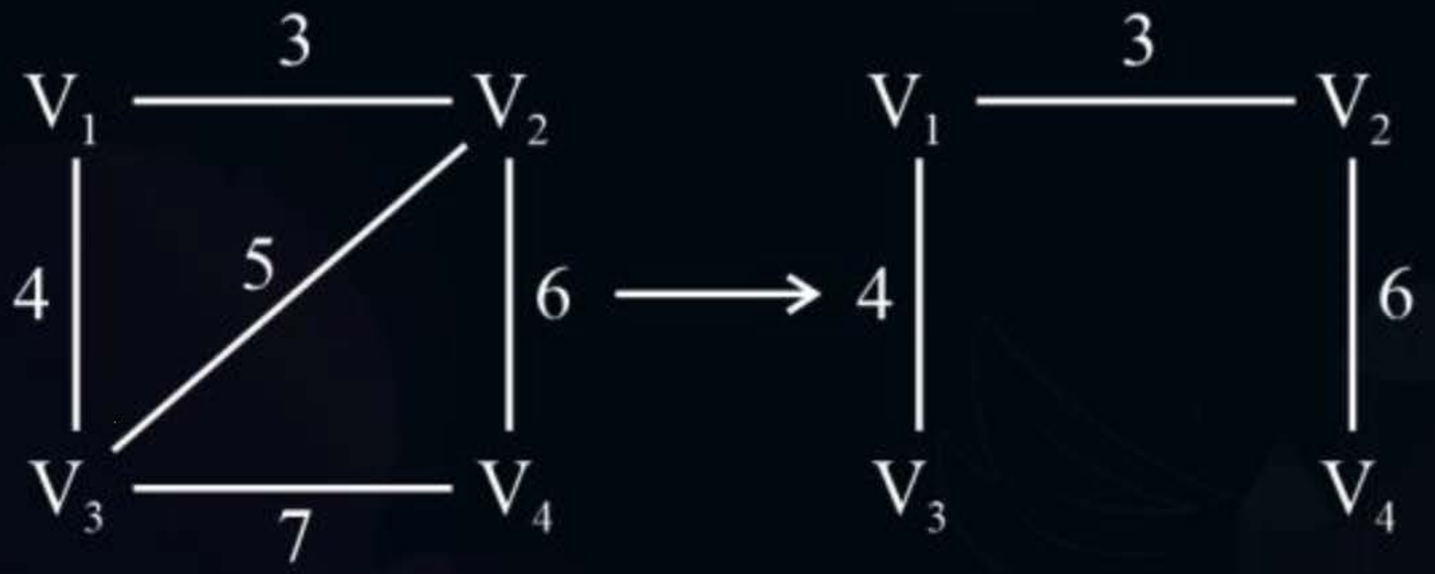


Ans :- 3+4=7



Topic : Greedy Algorithm

2) n = 4



n = 5

Ans. 3+4+6 = 13

General Pattern :-

$n = 4$

$\rightarrow 6$

$2(4-1)$

$$\text{Cost} = 3 + 4 + 6 + 8 \ldots\ldots$$

$$= 3 + 2(2 + 3 + 4 \ldots\ldots (n-1))$$

$$= 3 + 2 \times \left[\frac{n(n-1)}{2} - 1\right]$$

$$= 3 + n(n-1) - 2$$

$$= 1 + n^2 - n$$

$$= \boxed{n^2 - n + 1}$$

[MCQ]

Most Adv: and $1 \le y \le n$

#Q. Consider a graph whose vertices are points in a plane with integer co-ordinates (x, y) where $1 \le x \le n$, $n > 2$ is an integer. 2 vertices $\langle x_1, y_1 \rangle$ & $\langle x_2, y_2 \rangle$ are adjacent iff $|x_1 - x_2| \le 1$ and $|y_1 - y_2| \le 1$. The cost of such an edge is given by the distance between them. Compute the weight of min cost spanning tree of such graph for a given value of n.

n=3

44%

A n^2 → 9 ✗

B $(n^2 - 1) * 2$ → (9-1)×2 = 16 ✗

C $n^2 - 1$ = 8 ✓

D $n(n-1)$ → 3(3-1) = 6 ✗



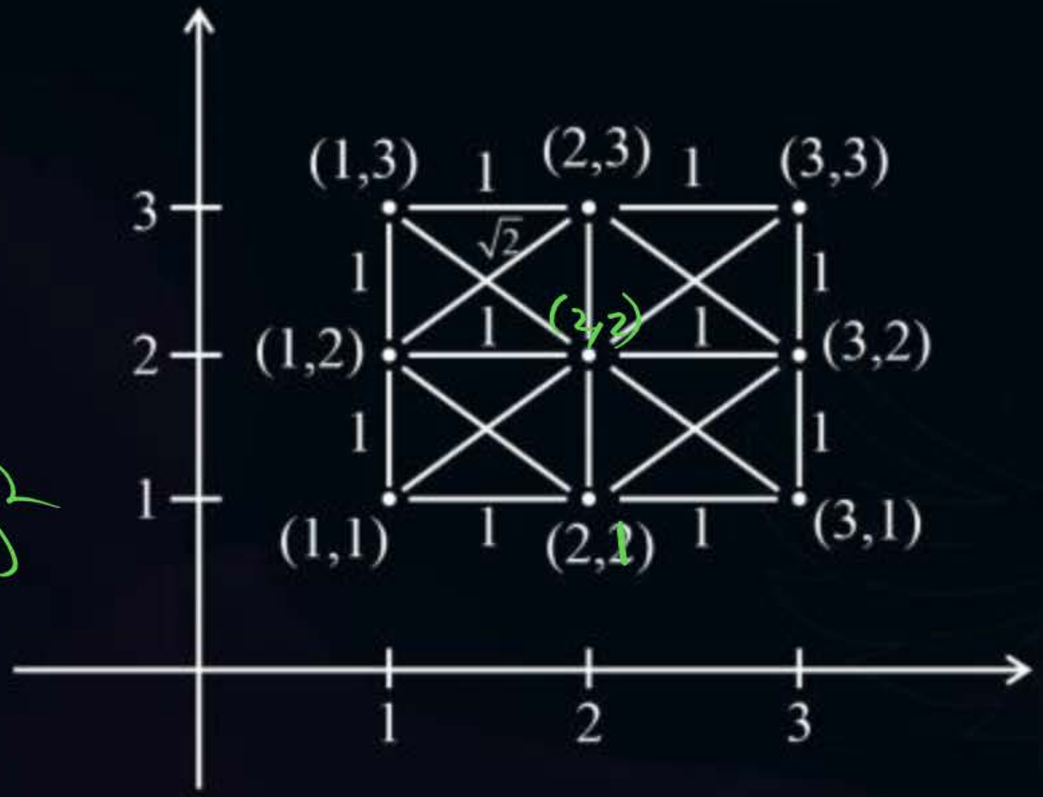
Topic : Greedy Algorithm

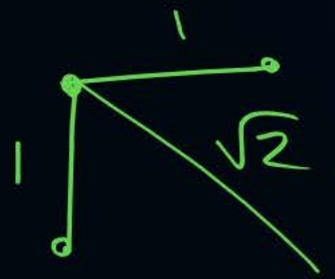
Solution:

n = 3

(x_1, y_1) (x_2, y_2)

$dist = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$

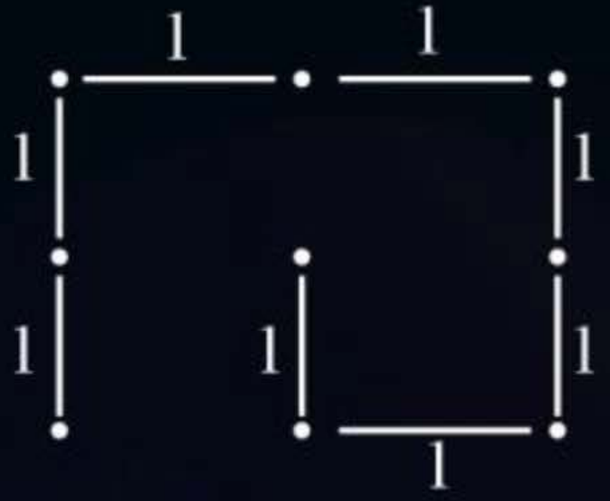






Topic : Greedy Algorithm

Kruskal



MCST

n

vertices $\rightarrow n^2$

$\downarrow$

MCST edges $\Rightarrow (n^2-1)$

every edge in MST $\rightarrow$ 1 $\rightarrow$ cost

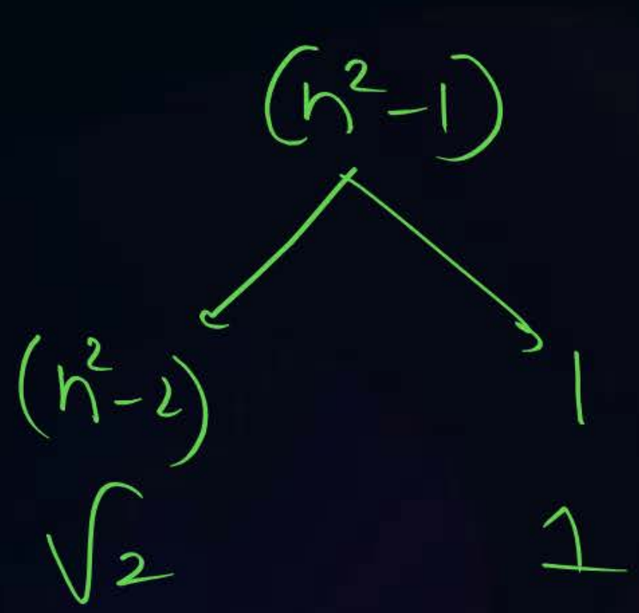
Total $= 1*(n^2-1) = (n^2-1)$

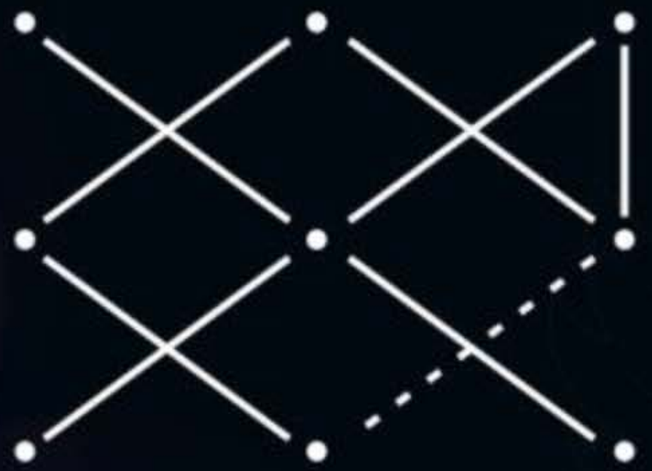


Topic : Greedy Algorithm

Variant 2: (Bonus)

#Q. What is the max cost possible for Spanning Tree?



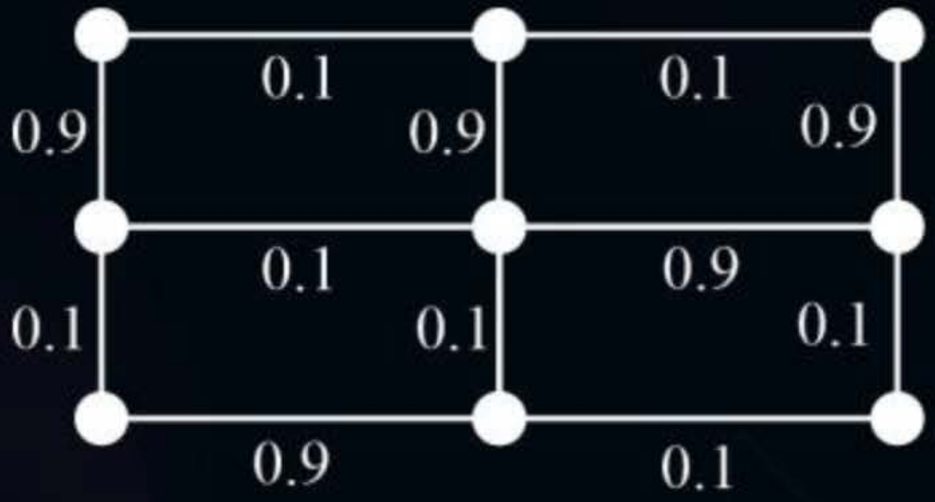


[NAT]

PYQ

#Q. Consider the following undirected graph with edge weights as shown

50%



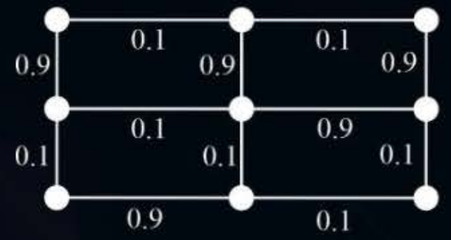
The number of minimum-weight spanning trees of the graph is _____.

count

Ans. 3

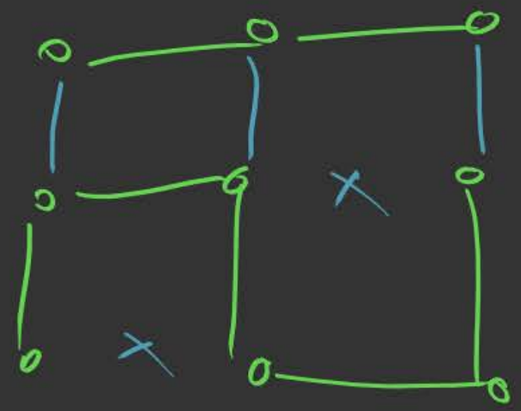
[NAT]

#Q. Consider the following undirected graph with edge weights as shown



The number of minimum-weight spanning trees of the graph is _____.

$^3C_1 = 3$

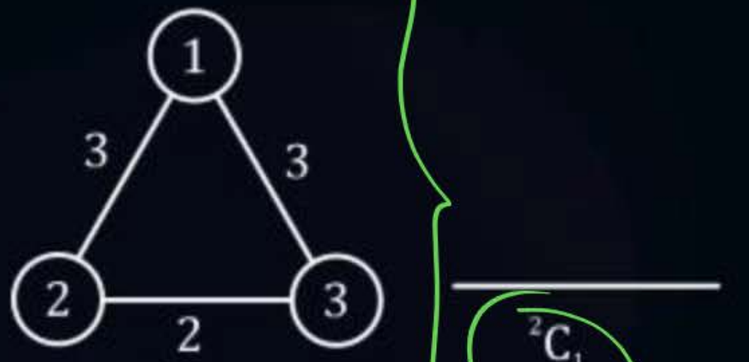


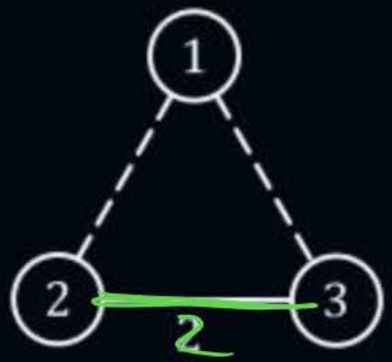


Topic : Greedy Techniques

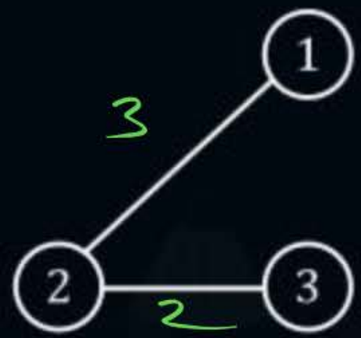
Concept:-

How many MCST?

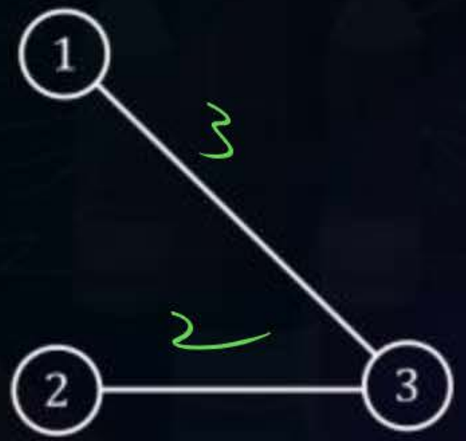




$^{2}C_{1}$
$= 2$



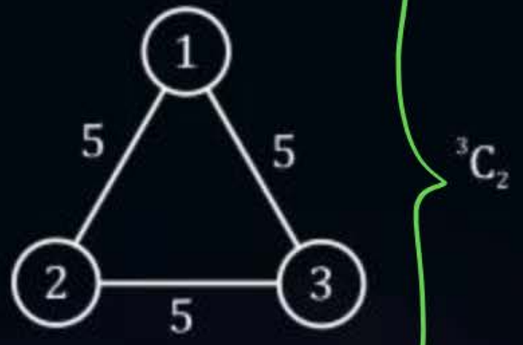
OR





Topic : Greedy Techniques

Eg.1.



$^3C_2$

$3C_2$

$\frac{3!}{2! \times 1!}$

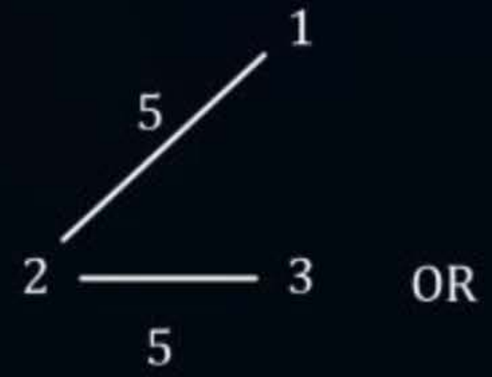
$= 3$

3. MCSTs

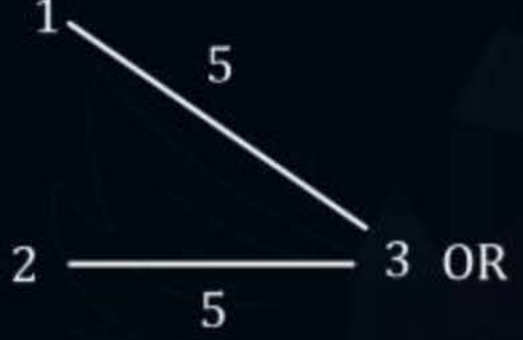
$n-1$

$= 2$

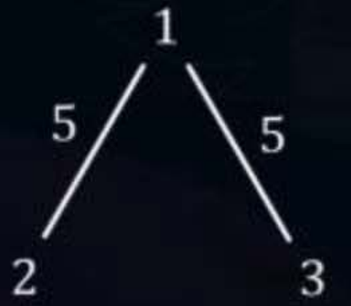
$3C_2$



OR



OR



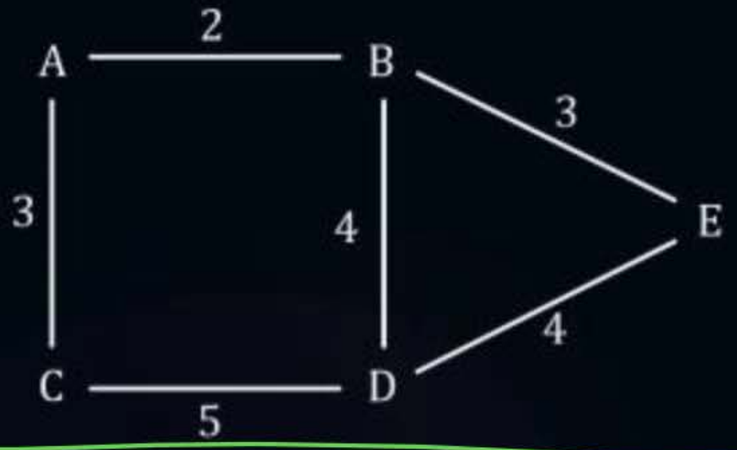


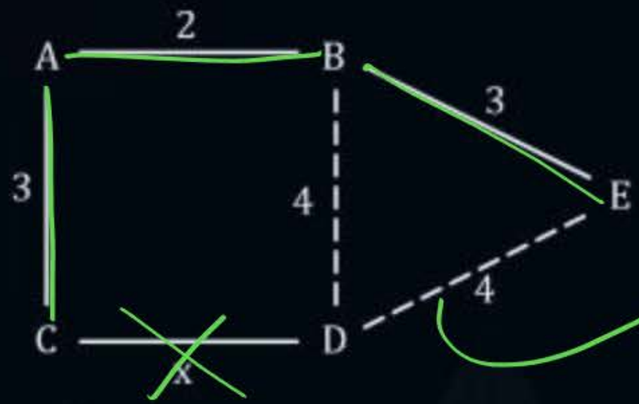
Topic : Greedy Techniques

Kruskal

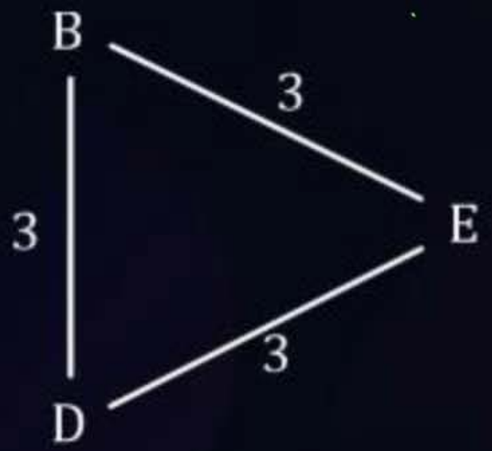
Eg.2.

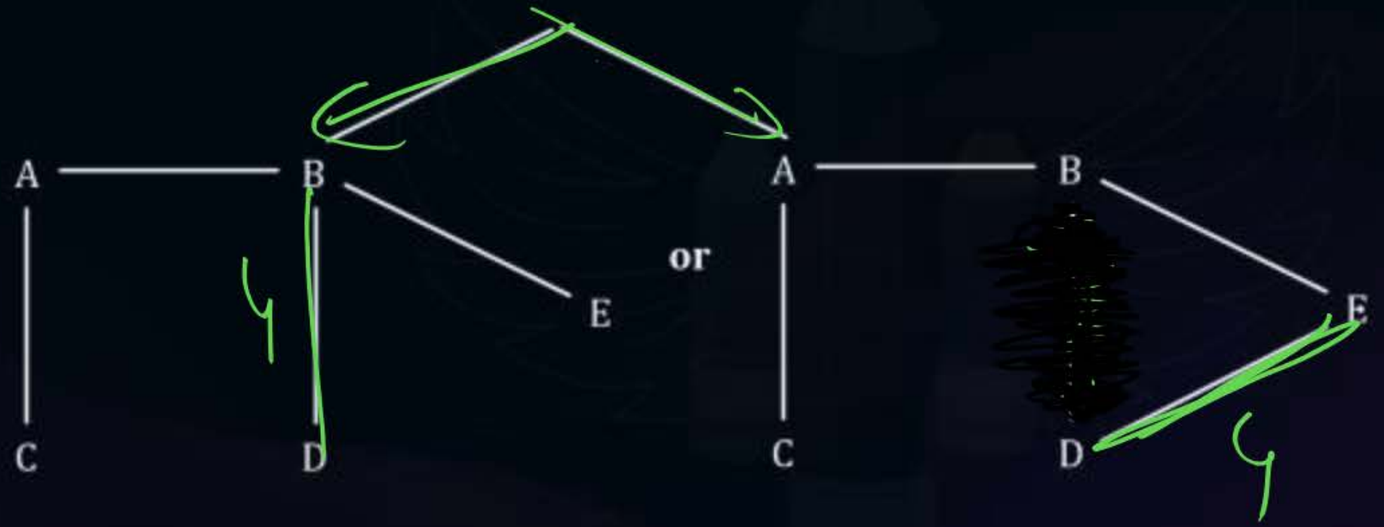
How many MCSTs?





$2C_1 = 2$







Topic : Greedy Techniques

PYQ

#Q. Let 'W' be the minimum weight among all edge weighs in an undirected connected graph. Let 'e' be a specific edge of weight 'w'. Which of the following is false?

i. There is a minimum spanning tree containing 'e' always → True

ii. Every minimum spanning tree has an edges of weight 'w'. → True

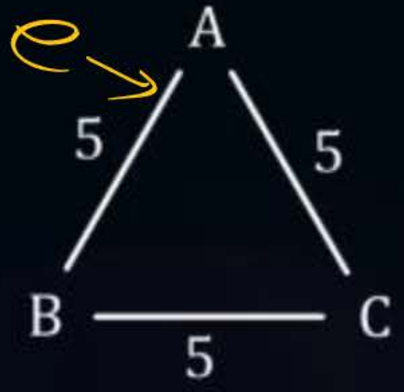
iii. 'e' is present in every minimum spanning tree. → False

iv. If 'e' is not present in a minimum spanning tree named 'T' then there will be a cycle formed by adding 'e' to T.



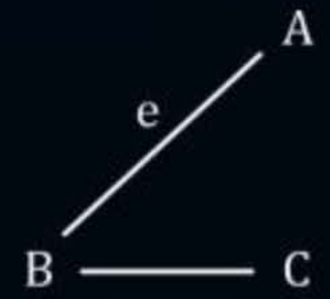
Topic : Greedy Techniques

Eg.3.



W = 5

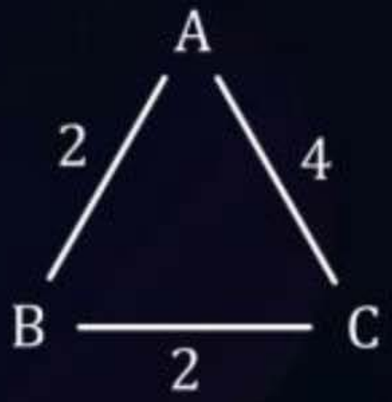
e = AB

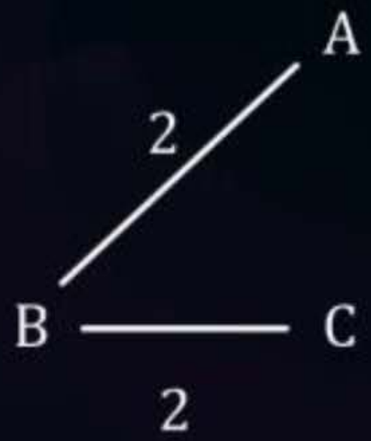


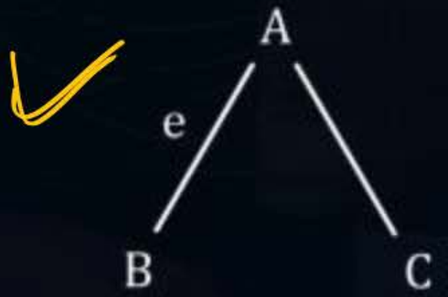


"e is not present"

Eg.4.





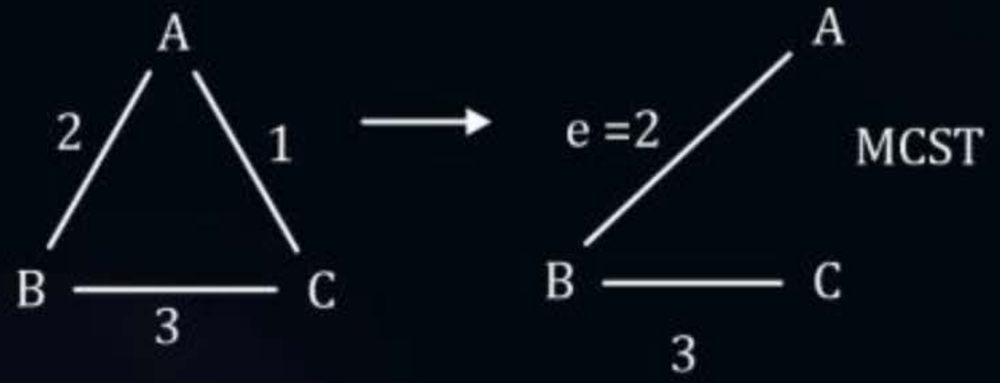




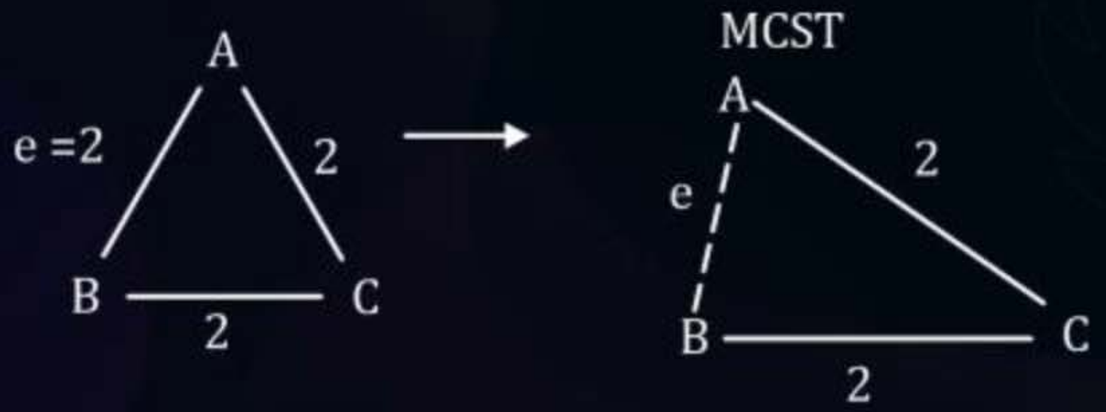
Topic : Greedy Techniques

H.W

Eg.5.



Eg.6.





Topic : Greedy Techniques

PYQ H.W

#Q. G= (V.E) is an undirected simple graph in each edge has a distinct weight, and e is particular edge of G. Which of the following statement about the minimum sapnning tree (MSTs) of G is/are True?

I. If e is the lighest edge of some cycle in G, then every MST of G includses.

II. If e is the heavist edge of some cycle in G, then every MST of G excluded e.

A. I only

B. II only

C. Both I and II

D. Neither I nor II



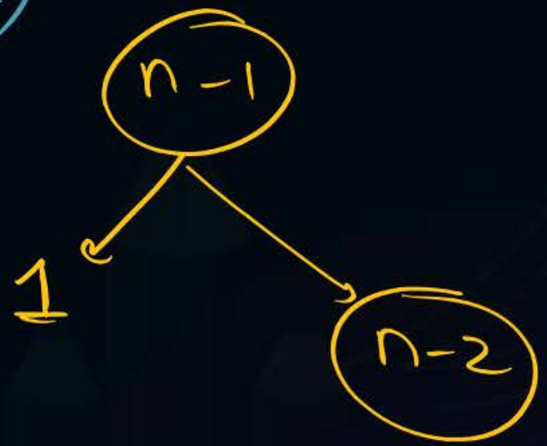
Topic : Greedy Techniques

$|E| = e$

$|V| = n$

temp array

```
1.  Algorithm Prims (E, cost, n, t)
2.  {
3.  Let (k,l) be an edge of minimum cost in E;        ⟶ O(e)
3.  mincost: cost[k,l]
4.  t[1,1]: =k; t[1,2]:=l;
5.  for i:=1 to n do // Initialize near.
6.  If (cost [i, l]< cost[i,k]) then near [i] := l;   ⟶ O(n)
7.  else near [i]: =k;
8.  near[k] := near [l] :=0;
9.  For i:=2 to n–1 do
```





Topic : Greedy Techniques

```
10.  { // Find n-2 additional edges for t.
11.  Let J be an index such that near [j] ≠0 and
12.  cost [j, near [j]] is minimum;
13.  t[I,1]:=j; t[i,2]: near [j];
14.  mincost:=mincost+cost[j,near [j]];
15.  near[j]:=0;
16.  for k:=1 to n do// Update near [].
17.  If ((near[k]≠0) and (cost [k, near[k]]>cost[k,j]))
18.  then near [k]:=j;
19.  }
20. Return mincost;
```

O(n)

j is not a part of MCST yet.

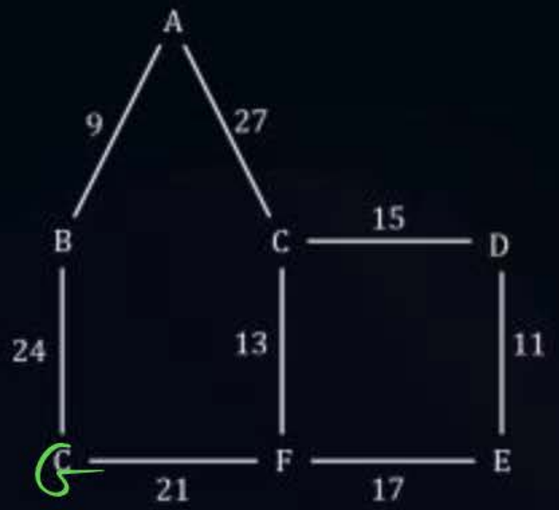
O(n)

O(n)

Topic : Greedy Techniques

walkthrough

Prims algo ~~although~~ & Explanation



A → C : 27 13

A → D : ∞ 15

A → E : ∞ 17 11

A → F : ∞

F → G : 21

B → G : 24



t[i, 1]

t[i, 2]



Topic : Greedy Techniques

$\min(k, l) \rightarrow (A, B) = 9$

t [1,1] = A

t[1,2] = B

After adding F: to MCST

C →F : 13

D → A : ∞ → D- C : 15

E → F: 17 → E-D : 11



Topic : Greedy Techniques

Minicost = 0

= 0 + 9 + 24 + 21 + 13 + 15 = 11

= 93



Topic : Greedy Techniques

Time complexity Analysis of Prime Algorithm;

1. Prve. Implementation (non-Heap based)

   $= e + n + c + n\,(n + c + n)$

   $= O(n^2)$

2. If heap is used in Prime algo implementation

   $TC = (n + e) \log n$

Case: If $e \approx O(n^2)$ then $n^2 \log n$



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

PYQs
Prims
TC



THANK - YOU